



Introduction to JSON

1. What is JSON
2. Comparing JSON with XML
3. JSON syntax
4. JSON parsing

1. What is JSON

- JSON, stands for JavaScript Object Notation
- Previously XML was being used as data exchange format, now-a-days it is replaced by JSON
- JSON is a text format for storing and transporting data
- It is self describing, hierarchical
- It is language and platform independent
- JSON parsing is easy than reading XML

2. Comparing JSON with XML

JSON	XML
Values have datatypes: number, string, boolean, array, object	Values have no datatypes
JSON supports only text and number values.	XML support various values such as number, text, images, charts, graphs, etc.
It doesn't support comments.	It supports comments.
JSON files are easy to read as compared to XML.	XML documents are relatively more difficult to read and interpret.
It is less secure.	It is more secure than JSON.

- It is easy for humans to read and write. It is easy for machines to parse and generate.
- It uses conventions that are familiar to programmers.
- A JSON is set of key-values pair
- Values can be string in double quotes, or a number, or true or false or null, or object or array.
- Delimiters used
 - i. Key and value separated by colon(:)
 - ii. key-value pair separated by comma(,)
 - iii. Strings are enclosed by double quotes(“)
 - iv. Arrays enclosed within square brackets([])
 - v. Objects enclosed within curly braces ({})
- Keys should be always in double quotes

- JSON object is very important object supported by all modern browsers.
- All properties and methods of JSON are static
- Parsing JSON string
 - The `JSON.parse()` is a static method which parses a JSON string according to the JSON syntax
 - It returns javascript object
 - Generally required when client receives JSON response from server
- Special cases for parsing
 - Parsing blank JSON generates error
 - Parsing null JSON string generates null
- Converting object into JSON
 - The `JSON.stringify()` is a static method which converts a JavaScript object to a JSON string.